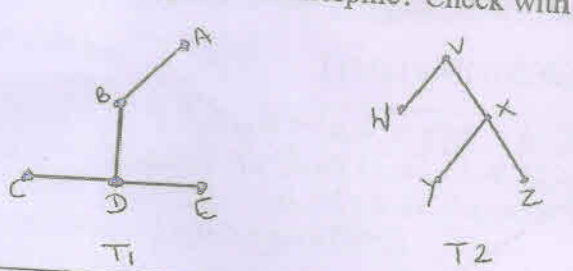
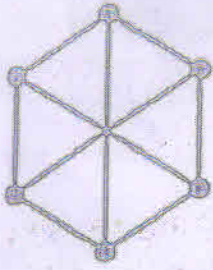


23.01.2023 (E) 1048

Maximum Marks: 100	Semester: August 2022 – December 2022 (Jan-23)	Examination: ESE Examination DSY (Reg + RT) Duration: 3 Hrs.
Programme code: 04	Class: SY	Semester: III (SVU 2020)
Programme: B.Tech.	Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: IT
Course Code : 116U04C301	Name of the Course: DAM	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	Define isomorphic trees Are following trees isomorphic? Check with justification	5
	 T_1 T_2	
ii)	Functions $f: R \rightarrow R$, $g: R \rightarrow R$ are defined as $(x) = 2x - 3$, $g(x) = 4 - 3x$ then find fog , fof , gof & $gogof$	5
iii)	let $A = \{1, 2, \dots, 8\}$. Let R be a relation defined as aRb if $a-b$ is multiple of 4. Write R as a set of ordered pairs and find matrix of R . If R is equivalence relation Find the partition of A induced by R	5
iv)	Find Laplace Transform of $\{e^{-5t}(\cos 6t + \sin 3t + 5\cosh 2t - 3\sinh 4t)\}$	5
v)	Obtain Fourier coefficient a_n in the Fourier series of $f(x) = x \sin(x)$ in $0 < x < 2\pi$ Hence find a_1 & a_2	5
vi)	Given any 6 integers from 1 to 10, show that some two of them have an odd sum	5
Q2 A	Solve the following	10
i)	Find $L\left\{5\frac{1}{t}e^{-2t}\sin 3t\right\}$	5
ii)	Find Inverse Laplace Transform of $\log\left(\frac{s+2}{s+3}\right)$	5
	OR	
Q2 A	Obtain Complex form of Fourier Series for $f(x) = e^x$ hence obtain Complex form of Fourier Series for e^{-x} & $\sinh(x)$ in $(-l, l)$	10
Q2 B	Solve any One	10

i)	Define Heaviside unit step function. Express the following function as Heaviside unit step function and find the Laplace transform $f(t) = \begin{cases} 0, & 0 < t < 1 \\ 2t, & 1 < t < 3 \\ 3t^2, & t > 3 \end{cases}$	10
ii)	Find Fourier series to represent $f(x) = 2x - x^2$ in $(0,3)$ Hence deduce that $\frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} \dots \dots \dots$	10
Q3	Solve any Two	20
i)	A tree T has k vertices of degree one, two vertices of degree two, three vertices of degree four and four vertices of degree three then find value of "k". Using the given Huffman tree decode the message (a) 00010001110011011101001 (b) 010011001111111	10
ii)	Using Laplace transform, solve the following differential equation $\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = 2(t^2 + t + 1), y(0) = 2, y'(0) = 0$	10
iii)	A = set of all positive integers, relation R on the set A is defined as $a R b$ iff $ a-b \leq 3$ Determine whether the relation R on a set A is reflexive, symmetric, transitive	10
Q4	Solve any Two	20
i)	State Convolution theorem and use it to find Inverse Laplace Transform of $\left\{ \frac{1}{(s^2+a^2)(s^2+b^2)} \right\}$	10
ii)	If $S = \{1, 2, 3, 4, 6, 12\}$, prepare table with respect to * in S where $a * b = \text{LC.M.}(a, b)$ Determine whether S is a monoid, a semigroup, a group with respect to * ?	10
iii)	Let $A = \{1, 2, 3, 4, 5\}$. Let R & S be an equivalence relations on A given as $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 3), (3, 3), (4, 4), (5, 5)\}$ $S = \{(1, 1), (2, 2), (3, 1), (1, 3), (3, 3), (4, 4), (5, 5)\}$ Write $R^{-1} \cap S^{-1}$, SOR	10

	find partition of A induced by $R^{-1} \cap S^{-1}$, SOR	
Q5	Solve any four	20
i)	Define Eulerian graph or Hamiltonian graph and check whether the given graph is Eulerian graph or Hamiltonian graph with justification? 	5
ii)	If $f: R - \{3\} \rightarrow R - \{0\}$ is defined as $f(x) = \frac{1}{x-3}$. Show that $f(x)$ is bijective and hence find $f^{-1}(x)$	5
iii)	Let $A = \{1, 2, 3, 4\}$, $B = \{a, b, c, d\}$, $C = \{x, y, z\}$ and let $R = \{(1, a), (2, d), (3, a), (3, b), (3, d), (3, c)\}$ be a relation from A to B and $S = \{(b, x), (b, z), (c, y), (d, z)\}$ be a relation from B to C. Write SOR Find Domain and Range of SOR	5
iv)	Evaluate $\int_0^{\infty} e^{-3t} t \sin h 2t dt$	5
v)	For half range cosine series of $f(x) = x(\pi - x)$, $0 \leq x \leq \pi$ If $a_0 = \frac{\pi^2}{6}$ and $a_n = \frac{-2(1+\cos n\pi)}{n^2}$ then using Parseval's Identity prove that $\frac{\pi^4}{96} = \frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4}$	5
vi)	Define zero divisors and unit elements of Z_n , hence find zero divisors and unit elements of Z_{14}	5

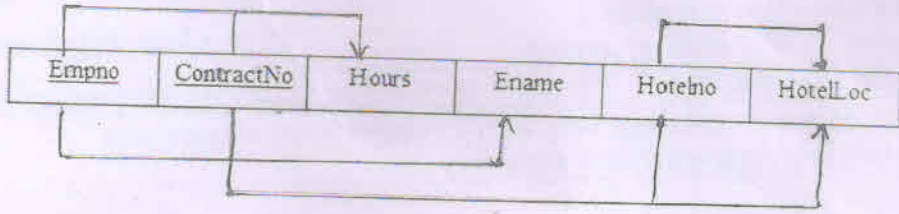


SOMAIYA
VIDYAVIHAR UNIVERSITY

Maximum Marks: 100		Semester: August 2022 – December 2022 – Jan- 2023	
Programme code: 04		Examination: ESE Examination DSY(Reg+KT) Duration: 3 Hrs.	
Programme: BTech in IT		Class: SY	Semester: III (SVU 2020)
Name of the Constituent College:		Name of the department: IT	
K. J. Somaiya College of Engineering			
Course Code: 116U04C303	Name of the Course: Database Management Systems		
Instructions: 1) Draw neat diagrams 2) All questions are compulsory			
3) Assume suitable data wherever necessary			

Que. No.	Question	Max. Marks
Q1	Solve any Four	
i)	List and explain any five significant differences between file processing and database management system.	20
ii)	What are the problems caused by redundancy?	5
iii)	Define ER model and describe the any four notations used in ER model with example.	5
iv)	List and explain any five advantages of DBMS.	5
v)	Explain the terms total participation and partial participation with example.	5
vi)	State the levels of abstraction in a DBMS.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	
i)	Explain security in database.	10
ii)	Explain various types of constraints in Database.	5
	OR	5
Q2 A	Explain the database system architecture in details with suitable diagram.	10
Q 2 B	Solve any One	
i)	Consider following database tables EMP(Eid, Ename, Salary , Address, Deptid) DEPT(Deptid, Dname,Address) construct the following SQL queries for this relational database. i) Find name of employee whose address is "Bandra" ii) Find name of employee earning highest salary iii) Find all employee working for "HR" department iv) Delete record of employee with minimum salary v) Display number of employees in each department	10
ii)	Construct an E-R diagram for a car insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents. Each insurance policy covers one or more cars, and has one or more premium payments associated with it. Each payment is for a particular period of time, and has an associated due date, and the date when the payment was received.	10

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	Consider the following dependency diagram of relation R and normalize till 3NF 	10
ii)	Consider the SQL query given below. a) Convert the following SQL query to Relational algebra expression b) Convert Relation algebra expression to query tree select P.Pnumber, P.Dnum, E.Lname, E.Address, E.Bdate from project as P, Department as D, Employee as E where P.Dnum = D.Dnumber and D.Mgr-ssn = E.ssn and P.Plocation = 'Mumbai'	10
iii)	Explain the dense and sparse indices with example.	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	Explain ACID properties of transaction.	10
ii)	Explain any two concurrency protocols in database systems.	10
iii)	What do you mean by serializable schedule? How would you test whether given schedule S is conflict serializable.	10

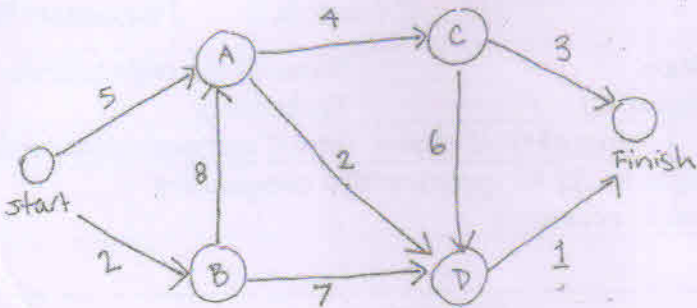
Que. No.	Question	Max. Marks
Q5	Solve on any four	20
i)	Write short notes on Primary key and Foreign key	5
ii)	Explain aggregate functions of DBMS.	5
iii)	Describe View with example.	5
iv)	Explain Transaction state diagram.	5
v)	Write notes on mapping of cardinality with suitable diagram.	5
vi)	Explain shadow Paging with diagram.	5

1.2.2023(E)


SOMAIYA
 VIDYAVIHAR UNIVERSITY

Semester: August 2022 – December 2022 / JAN 2023		
Maximum Marks: 100	Examination: ESE Examination DSY (Reg+KT) Duration: 3 Hrs.	
Programme code: 04	Class: S.Y	Semester: III(SVU 2020)
Programme: B.Tech		
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: Information Technology
Course Code: 116U04C305	Name of the Course: Data Communication and Networking	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	What is the physical path and channel in network?	5
ii)	What is a network topology? What are its types? Which is mostly used cabled network topology especially in large businesses?	5
iii)	Enlist the networking devices with its role.	5
iv)	State the differences between TCP and UDP protocol	5
v)	What are the advantages of dividing an Ethernet LAN with a bridge?	5
vi)	Compare Random access protocol with Controlled access protocol?	5
Q2 A	Solve the following	10
i)	What is role of presentation layer in OSI?	5
ii)	Comparison between ISO /OSI and TCP/IP Model?	5
OR		
Q2 A	What are the different layers of TCP/IP? What are the functions of each layer? Which are the different protocols working at each layer?	10
Q 2 B	Solve any One	10
i)	What is Domain Name System (DNS)? Briefly explain flat name space and hierarchical name space with diagrams. Explain the need of DNS in the IP networks?	10
ii)	Explain TCP/IP Header with suitable diagram.	10
Q3	Solve any Two	20
i)	An organization with given block 200.16.178.0/16. The administrator wish to give IP address so that he will creates 16 subnets. Find its subnet mask? Find the number of addresses in each subnet? Find the first address and last address of the block?	10

ii)	What is a routing? What are the cases considered while choosing the path to reach single destination from multiple path? How Dijkstra's algorithm finds shortest path for a given network? 	10
iii)	What are the various services offered by network layers?	10
Q4	Solve any Two	20
i)	Given data word 110000101011 and divisor 1011. Then show the generation of code word at a sender site (using binary division). Show checking of code word at the receiver's site. Assume no error.	10
ii)	Compute the checksum at sender and receiver side? Consider the data unit to be transmitted over network is- 10011001111000100010010010000100. Consider 8 bit checksum to be used.	10
iii)	What are the differences between flow control mechanism in data Link Layer and Transport Layers? Explain Working of Stop and wait protocol. State its 2 disadvantages.	10
Q5	(Write notes / Short question type) on any four	20
i)	Describe Go back -N ARQ with an example.	5
ii)	Explain the concept of switching in data communication along with its different categories.	5
iii)	How does a single-bit error differ from a burst error?	5
iv)	What is a HDLC protocol? What are its features ?	5
v)	Explain sliding window protocol with suitable diagram.	5
vi)	What is classful addressing? How it differs from classless addressing?	5



Semester: August 2022 – December 2022 January 2023	
Maximum Marks: 100	Examination: ESE Examination DSV(Reg+KT) Duration: 3 Hrs.
Programme code: 04	Class: SY
Programme: B.Tech IT	B.Tech
Semester: III (SVU 2020)	
Name of the Constituent College:	Name of the department: IT
K. J. Somaiya College of Engineering	
Course Code: 116U04C304	Name of the Course: Digital Systems
Instructions: 1) Draw neat diagrams 2) All questions are compulsory	
3) Assume suitable data wherever necessary	

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	Convert the following to binary and then to gray code i) $(1111)_{16}$ ii) $(BC54)_{16}$ iii) $(237)_8$ iv) $(164)_{10}$ v) $(323)_8$	5
ii)	Find the minterms for the function $f(a,b,c,d) = a'(b'+d) + acd'$	5
iii)	Define Decoder. Write truth table for 3 to 8 line Decoder and draw block diagram.	5
iv)	Construct JK Flip-Flop by using SR Flip-Flop and NAND Latch.	5
v)	What are the actions taken by CPU when an instruction fetched from memory?	5
vi)	Explain any 5 register from the group of general purpose and index registers of 8086 microprocessor.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	Draw and explain working of 4-bit Johnson Counter. Draw suitable Timing Diagram.	5
ii)	Explain the working of Serial-In-Parallel-Out Shift Register. Draw Timing Diagram if Input Data = 1011	5
Q2 A	Convert D Flip-Flop into JK Flip-Flop. Design block diagram of JK FF using D FF.	10
Q 2 B	Solve any One	10
i)	Design 3-bit Synchronous UP counter using T Flip-Flop.	10
ii)	Design 3-bit Synchronous DOWN counter using D Flip-Flop.	10

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	Using 4:1 Multiplexer implement the function $f(a,b,c,d) = \sum(0, 4, 5, 10, 11, 12, 15)$	10
ii)	Design Full Adder using Decoder and OR gates.	10
iii)	What is Magnitude Comparator? With the help of suitable truth table and K-Map design 2-bit Comparator using logic gates.	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	
i)	Explain Basic Instruction Cycle with state diagram.	20
ii)	What is RAM memory? Explain the working of DRAM cell with neat diagram. Give any 4 difference between SRAM and DRAM.	10
iii)	Explain the Bus Interconnection scheme of Von Neumann Architecture.	10

Que. No.	Question	Max. Marks
Q5	(Write notes/ Short question type) on any four	
i)	Rewrite and Implement the given expression using NAND Gate only $F = A'D + AD'(B+C')$	20
ii)	What is Encoder? Design an Octal to Binary Encoder using logic gates.	5
iii)	Draw Truth Table, Characteristic Equation and Excitation Table of D Flip-Flop.	5
iv)	<u>Write a note on design and characteristics of memory hierarchy.</u>	5
v)	How to write ADD instruction in five different addressing modes and specify location of operand which is placed in memory.	5
vi)	Explain is Memory Segmentation in 8086 microprocessor.	5